

Day 1 — Reducing LLM Hallucinations via Structured Knowledge Integration

LLM Research Log | Souvik Pal

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Background

I started this research with no prior background in LLMs, beyond general awareness of tools like ChatGPT. During JAVA Lab sessions, Dinesh Sir explained how large language models function and, in doing so, introduced **hallucination** — the tendency of LLMs to generate fluent but factually incorrect output — as one of the field's central open problems.

This raised a natural question: how do these models actually work under the hood, and what real architectural gaps cause hallucination? Sir outlined the scope for research in this space and provided initial resources covering Knowledge Graphs (KG), Meta-Knowledge Graphs (MKG), GraphRAG, KG-Adapters, and Structured Reasoning as active approaches to this problem. This log documents my own study path through these topics.

Reducing LLM Hallucinations via Structured Knowledge Integration

Objective

To understand the root causes of hallucination in LLMs and evaluate how Knowledge Graphs and neuro-symbolic AI approaches address them — with a specific look at practical frameworks (the NIT Durgapur pipeline and the Meta-Knowledge Guided / MKG framework).

The Problem

Traditional LLMs hallucinate because of three compounding architectural issues:

- **Implicit knowledge storage** — facts are distributed across billions of floating-point weights with no way to directly verify them
- **Probabilistic generation** — the model predicts the most probable next token, not the most factual one
- **Open-loop architecture** — there is no internal verification or fact-checking step after generation

A common misconception is that scaling model size alone solves this. In reality, hallucination is a structural limitation that persists even in larger models — the fix requires redesigning the architecture to include external verification loops, not just adding parameters.

Methodology

Knowledge Graphs (KG) store facts as discrete, queryable triples — (Head Entity) → [Relation] → (Tail Entity) — making them traceable and verifiable, unlike knowledge implicitly encoded in model weights.

Neuro-Symbolic AI combines the language fluency of neural models (LLMs) with the factual precision of symbolic systems (Knowledge Graphs), aiming to get the strengths of both while offsetting each one's individual weaknesses.

I studied this through two frameworks:

1. NIT Durgapur — 4-Phase Pipeline, addressing four research questions:

Phase	Focus	Outcome
RQ1	Ontological Alignment — converting LLM output into KG-compatible triples via adapter layers	31% reduction in hallucination
RQ2	Self-Verification Loop — a Generate → Validate → Refine cycle	Closed-loop correction (stronger than standard RAG, which only retrieves without correcting)
RQ3	Uncertainty Estimation — verifying only when the model is uncertain	Reduces unnecessary computation
RQ4	Conflict Resolution — resolving disagreements using Bayesian trust scoring	94% accuracy on multi-hop reasoning tasks

2. Meta-Knowledge Guided (MKG) Framework — introduces the idea of *meta-knowledge*: not just facts, but knowledge about how to use and verify facts (analogous to a student double-checking their work, not just knowing the textbook). MKG is built from three cooperating components:

- **Meta-Executor (Me)** — generates initial responses from retrieved facts
 - **Meta-Assessor (Ma)** — evaluates outputs for hallucination/consistency on a 1–5 scale
 - **Meta-Insight Processor (Mip)** — generates corrective feedback and learns patterns over time
- MKG also retrieves knowledge using three complementary methods — BGE semantic embeddings, ColBERT late-interaction, and BM25 lexical retrieval — fused via Reciprocal Rank Fusion, so that abstract queries, precise phrases, and exact keyword matches are all covered. It further separates memory into a Short-Term Memory (session-specific context) and a Long-Term Memory (a persistent vector store of reusable meta-knowledge across sessions).

An important real-world caveat I noted: knowledge graphs themselves are not perfectly clean — they typically contain 5–15% inaccurate, outdated, or contradictory data. Blindly trusting external KG data is therefore just as risky as blindly trusting the LLM's internal weights. MKG addresses this via Bayesian trust scoring, temporal validity stamps, and consensus/voting mechanisms for conflicting sources.

Results

- RQ1's adapter-layer alignment alone reduced hallucination by **31%**

- RQ4's conflict-resolution approach reached **94% accuracy** on multi-hop reasoning tasks
- On the GrailQA benchmark, MKG improved accuracy by **+30% over baseline LLMs**, **+17% over standard RAG**, and **+5.4% over other state-of-the-art methods**

Conclusion

Hallucination is best understood as an architectural problem, not an inherent flaw that scale alone can fix. Structured knowledge (via Knowledge Graphs) combined with self-verification loops and meta-knowledge learning offers a measurable, benchmarked path toward more trustworthy LLMs.

Possible Future Directions (noted for later study)

- Domain-specific adapters (medical/legal/finance)
- Temporal knowledge decay — auto-expiring outdated facts
- Explainability audit trails for high-stakes domains
- Multi-modal KG integration (text + images)
- Real-time KG updates via streaming pipelines

Personal Reflection

This was our very first day on this research. I didn't know anything about this topic before — Dinesh Sir told us all the basic things about LLMs, and the future scope of this field, what I could work on and what I had to do next. Later Sir gave us resources to study. On this first day I learned the basics of what LLM, KG, and MKG actually are. At first it looked complex, but after going through it properly, I got interested in it, and that's when I started focusing more seriously on the research questions.